

Dieckow 2024 · 10-guild gLV · $N = 10$ patients · Nishioka (2026)

Guild-Level Dynamical Analysis

Keystone · Network · Permutation · Tipping · Relapse · Prevention · 3D Landscape

Periodontitis relapse is governed by the topology of microbial interaction networks — not merely by bacterial growth rates. Prevention thresholds and relapse timing can be mathematically derived from patient-specific gLV dynamics.

Key message

Background & Mathematical Model

Dieckow et al. 2024 · 10-guild gLV · N = 10 patients (CT1 commensal / CT2 dysbiotic) · 4 time-points

Clinical Background

~700 species live in the oral cavity,
organised into 10 functional guilds.

Healthy state (CT1)

Actinobacteria, Bacilli dominant
GDI < 0 (commensal)

Dysbiotic state (CT2)

Bacteroidia, Fusobacteria expand
GDI > 0 (dysbiotic)

Clinical question:

Why do some patients relapse after
treatment while others do not?

Dataset: 10 patients, 4 time-points (wk 0/1/3/6)

Generalised Lotka-Volterra (gLV)

$$d\varphi_i/dt = \varphi_i \cdot (b_i + \sum_j A_{ij} \varphi_j)$$

- φ_i relative abundance of guild i
- b_i intrinsic growth rate (patient-specific)
- A_{ij} interaction j → i (+ mutualism, – competition)
- GDI $\log(\varphi_{\text{dys}}) - \log(\varphi_{\text{com}}) < 0 = \text{commensal}$

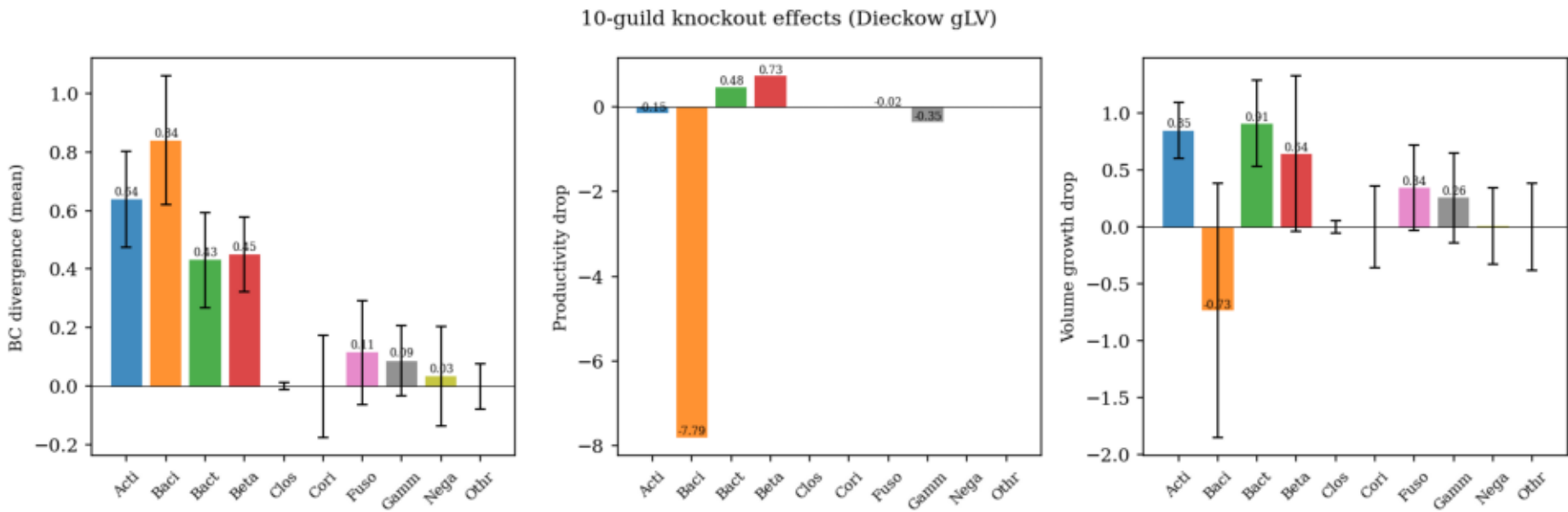
Fitting:

L-BFGS-B minimises prediction MSE
→ A (10×10) and b (10 × 10)
A encodes ecological network → long-term fate

Fitting: L-BFGS-B minimises MSE over 4 time-points → recovers A (10×10) and b (10 patients × 10 guilds). | All 6 downstream analyses are derived from the single fitted A matrix.

Guild-Level Keystone Analysis

Knockout: remove one guild → simulate 21 d → measure Bray-Curtis shift / volume change



Compositional keystone:

Bacilli BC = 0.839

removal shifts community most

Competitive suppressor:

vol_drop = -0.73

other guilds fill the space

Growth engine:

Actinobacteria +0.846

drives total biomass

Strongest interaction:

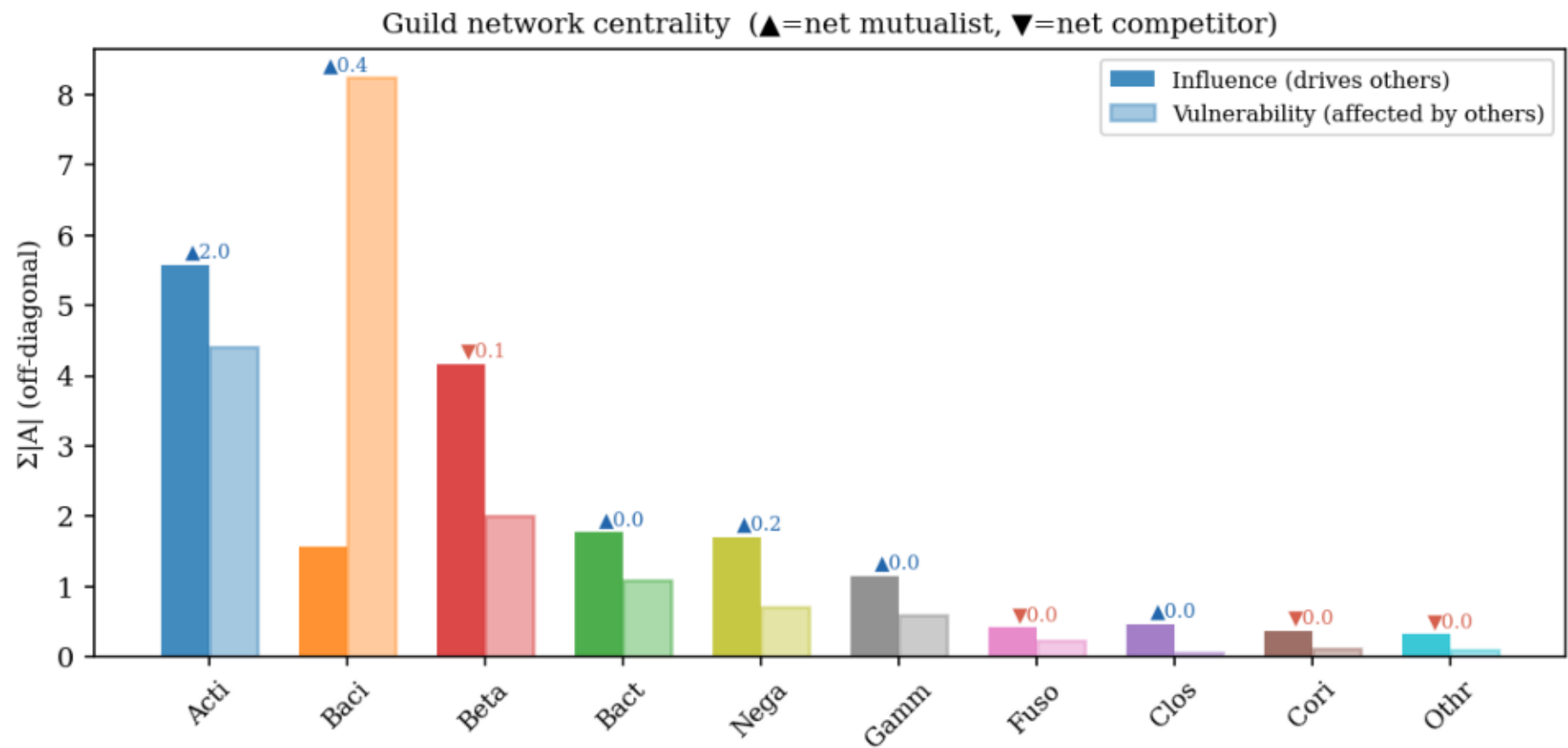
Acti → Baci A = +3.43

Acti stimulates Baci

Bacilli: BC divergence 0.839 (highest), vol_drop -0.73 → compositional keystone & competitive suppressor. | Actinobacteria: vol_drop +0.846 → growth engine. | Strongest off-diagonal interaction: Acti → Baci A = +3.43.

Guild Network Centrality: Influence, Vulnerability, Net Role

Influence = $\sum |A_{ji}|$ · Vulnerability = $\sum |A_{ij}|$ · Net = $\sum A_{ij}$ · LOO sign consistency across 10 folds



Mutualist hub:

Actinobacteria

influence = 5.58 (highest)

net effect = +2.02

LOO sign = 1.0 (stable)

Most regulated:

Bacilli

vulnerability = 8.24

controlled by others

Net competitor:

Betaproteobacteria

net effect = -0.14

Acti-Baci-Beta triad

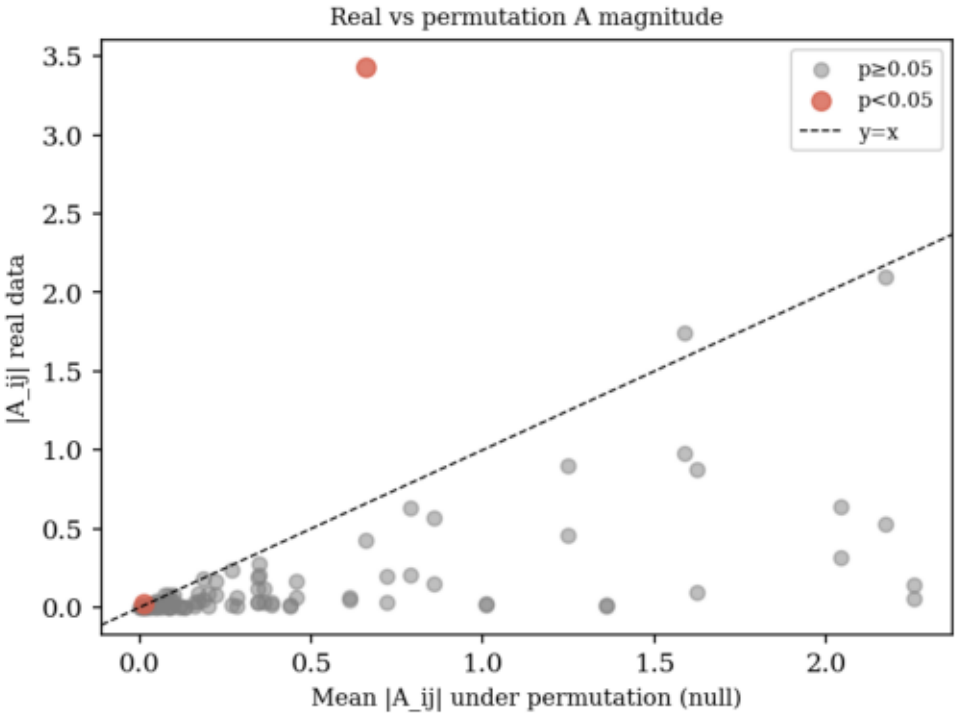
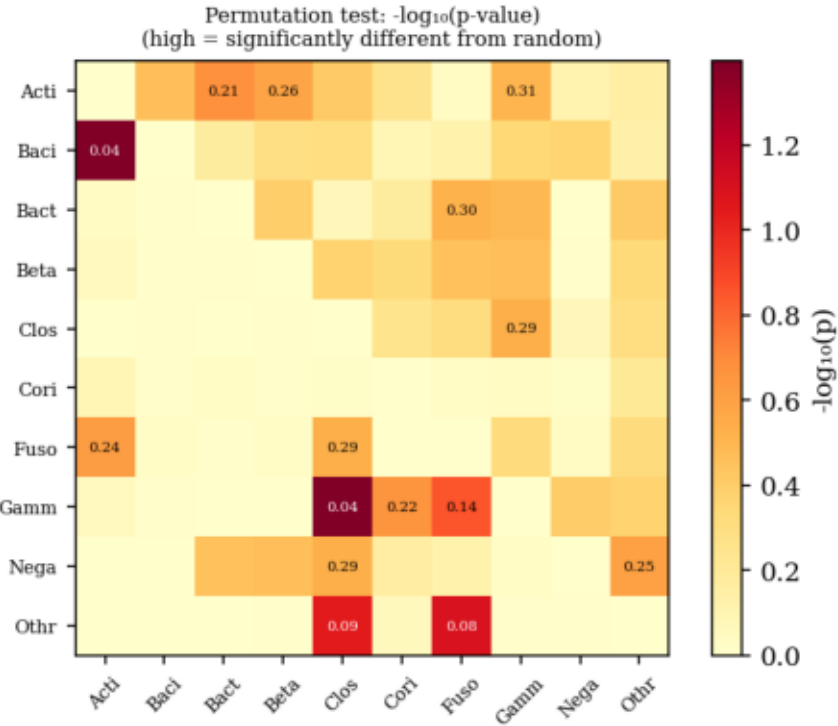
stabilises dysbiotic attractor

Actinobacteria: influence 5.58, net +2.02 → mutualist hub. Bacilli: vulnerability 8.24 → most regulated. Betaproteobacteria: net -0.14 → only net competitor. LOO: Acti→* sign consistency = 1.0 (all folds agree).

A Matrix Statistical Validation: Permutation Test (n = 100)

Null: shuffle patient labels → refit A via L-BFGS-B → p-value = fraction of $|A_{perm}| \geq |A_{real}|$ (24-core parallel)

Permutation test (n=100 permutations) 2/90 off-diag pairs significant (p<0.05)



Significant ($p < 0.05$):

2 / 90 pairs

Baci → Acti

$A = +3.43$, $p = 0.040$

strongest real interaction

Gamm → Clos

$A = -0.023$, $p = 0.040$

LOO sign stability:

Acti→* SC = 1.0 (stable)

Gamm→* SC ≈ 0 (limited)

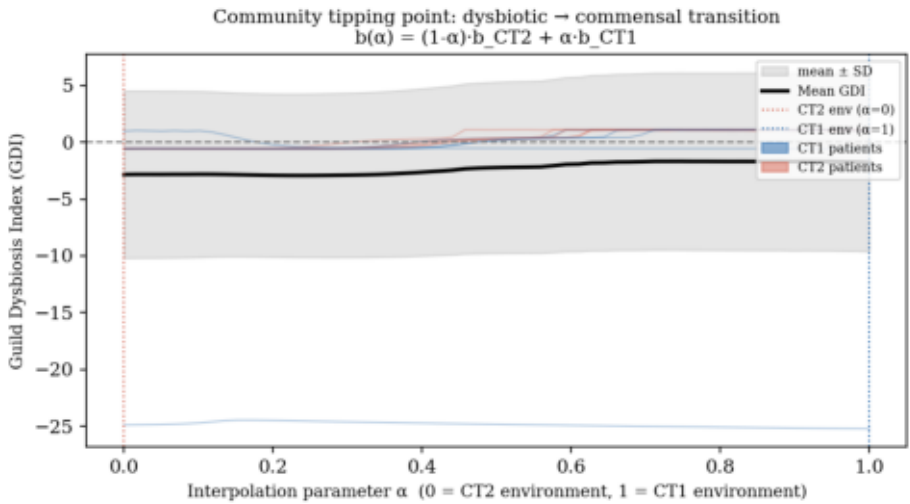
N = 10 limits power;

structure is meaningful

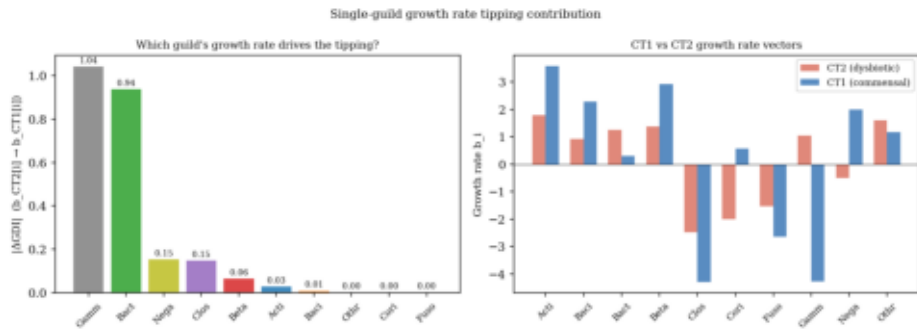
2/90 pairs significant at $p < 0.05$: Baci→Acti ($A = +3.43$, $p = 0.040$) and Gamm→Clos ($p = 0.040$). | LOO confirms Acti→* sign stability (SC = 1.0 all folds). | Low significance consistent with N = 10; pattern reflects structure, not noise.

Tipping Point Analysis: CT2 → CT1 b-Environment Transition

Left: $b(\alpha) = (1-\alpha) \cdot b_{CT2} + \alpha \cdot b_{CT1}$ | Right: sweep one b_i while holding others at CT2 mean



(a) α -interpolation CT2 → CT1 environment



(b) Single-guild b sweep: $|\Delta GDI|$ effect size

Alpha scan:

No GDI = 0 crossing

single commensal attractor

A matrix dominant

Top single-guild drivers:

Gammaproteobact.

$\Delta GDI = 1.04$ (#1)

$\Delta b = -5.31$

Bacteroidia

$\Delta GDI = 0.94$ (#2)

$\Delta b = -0.96$

Equilibrium GDI:

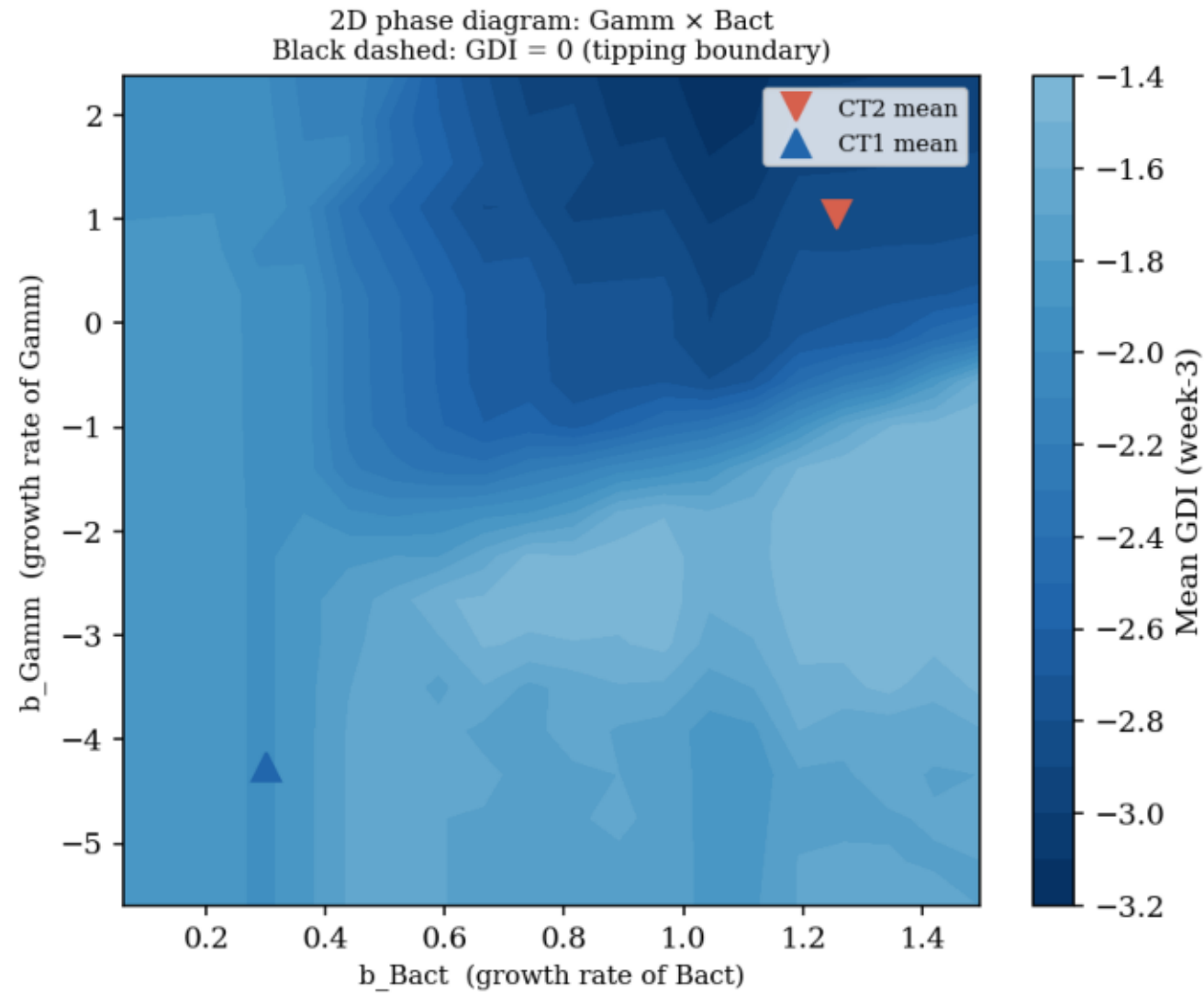
Patient D: -0.48 (commensal)

All others: > 0

No α^* crossing: $GDI < 0$ for all $\alpha \rightarrow$ A matrix drives commensalism regardless of b environment. | Single-guild drivers: Gammaproteobacteria ($\Delta GDI = 1.04$) and Bacteroidia ($\Delta GDI = 0.94$). $\rightarrow$ Structural attractor: interventions must alter A (ecology), not just b .

2D Phase Diagram: Gammaproteobacteria × Bacteroidia

20 × 20 grid · all other b fixed at CT2 mean · week-3 GDI · dashed = GDI = 0 tipping boundary



Tipping boundary:

GDI = 0 (dashed)

visible in 2D space

Commensal zone:

low b_Gamm + low b_Bact

Patient positions:

CT2 (▼): dysbiotic zone

CT1 (▲): near boundary

Clinical strategy:

Jointly reduce b_Gamm

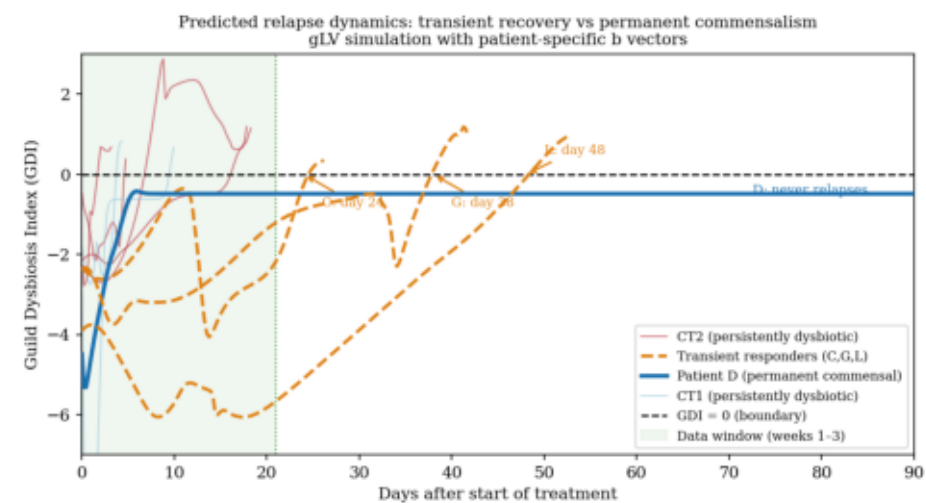
and b_Bact for efficient

transition to commensal state

Top-2 tipping guilds (Δ GDI 1.04 and 0.94). Commensal region at low b_Gamm and low b_Bact. CT2 (▼) deep in dysbiotic zone; CT1 (▲) near boundary. → Joint reduction of Gamm and Bact growth rates is the most efficient ecological intervention.

Predicted Relapse Dynamics: Three Patient Archetypes

gLV simulation $t = 150 \text{ d}$ · patient-specific b vectors · $GDI = \log(\varphi_{\text{dys}}) - \log(\varphi_{\text{com}})$



(a) GDI trajectories to 90 d

(b) Week-3 GDI as prognostic marker

① Persistent dysbiotic

A, B, E, F, H, K

GDI > 0 from day 1

② Transient responder

C → day 24

G → day 38

L → day 48

commensal at wk-3,

attractor pulls back

③ Permanent commensal

Patient D only

GDI stable < 0

Prognostic rule:

GDI(wk3) ↑ → earlier relapse

① Persistent dysbiotic (A,B,E,F,H,K): GDI > 0 throughout. ② Transient responders (C→24d, G→38d, L→48d): commensal at wk-3, relapse by day 50. ③ Permanent commensal (D only): stable GDI < 0 at $t = 150\text{d}$. → Week-3 GDI is a prognostic marker for relapse timing.

Prevention Threshold & External Validation (Joshi 2025)

Left: min. Δb_i to prevent relapse · Right: model eq. GDI vs Joshi 2025 peri-implant cohort (N = 127)

Prevention:

Patient C:

$\Delta b_{\text{Bact}} = 1.33$ sufficient

→ permanent commensal

Patients G & L:

no single-guild threshold

deep dysbiotic attractor

Joshi validation:

Predicted eq. GDI:

+0.65 (model)

Joshi PI mean:

+0.53 (data)

9/10 patients match

3-wk data predicts

long-term dysbiosis

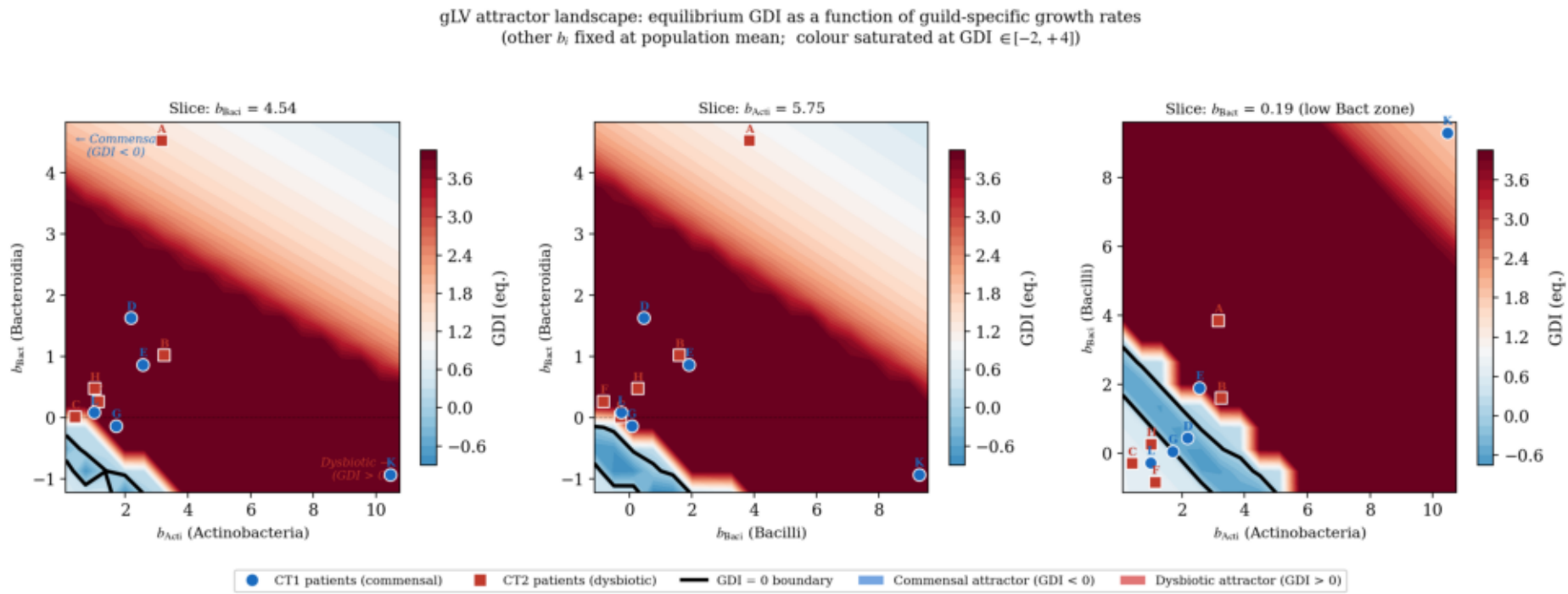
(a) Required Δb per guild

(b) Joshi 2025 peri-implant cohort

Prevention: Patient C convertible with $\Delta b_{\text{Bact}} = 1.33$; G & L require multi-guild change → deep attractor. | Joshi validation: predicted eq. GDI +0.65 $\approx$ Joshi PI mean +0.53 (9/10 patients; $\Delta = 0.12 < \text{SD} = 1.30$).

3D Attractor Landscape: GDI = 0 Boundary in Growth-Rate Space

18³ grid · Acti × Baci × Bact · t = 80 d · 12-core parallel · blue = commensal | red = dysbiotic



Dysbiotic dominance:

98% of grid: GDI > 0

Only 2% commensal

Critical threshold:

$b_Bact < \sim 1.5$

→ commensal equilibrium

Bacteroidia = dominant

control variable

Patient positions:

CT1: near boundary

CT2 patient A ($b=4.5$):

deep in dysbiotic zone

Must lower b_Bact ,

not just change $\phi(0)$

98% of parameter space is dysbiotic. Commensal region confined to $b_Bact < \sim 1.5$ — Bacteroidia growth rate is the dominant control variable. CT1 patients near boundary; CT2 patient A ($b_Bact = 4.5$) deep in dysbiotic zone. → Attractor robustness is structural, not initial-condition sensitive.

Conclusion

Periodontitis relapse is governed by the topology of microbial interaction networks, not merely by bacterial growth rates. Prevention thresholds can be mathematically derived.

- ① Bacilli = compositional keystone ($BC = 0.839$); Actinobacteria = growth engine.
- ② Baci $\rightarrow$ Acti ($A = +3.43$, $p = 0.040$) is the sole statistically validated interaction.
- ③ A matrix creates a single commensal attractor — b alone cannot flip the community state.
- ④ Three archetypes: persistent dysbiotic / transient responder / permanent commensal (D only).
- ⑤ Patient C preventable with $\Delta b_{\text{Bact}} = 1.33$; 98% of growth-rate space is dysbiotic.
- ⑥ Validated externally: model eq. $GDI + 0.65 \approx \text{Joshi 2025 peri-implant mean } +0.53$.